

Simulation and Analysis of pp Collisions at $\sqrt{s} = 13.6$ TeV

Tomasz Pawlaczyk

CERN Internship Recruitment Task

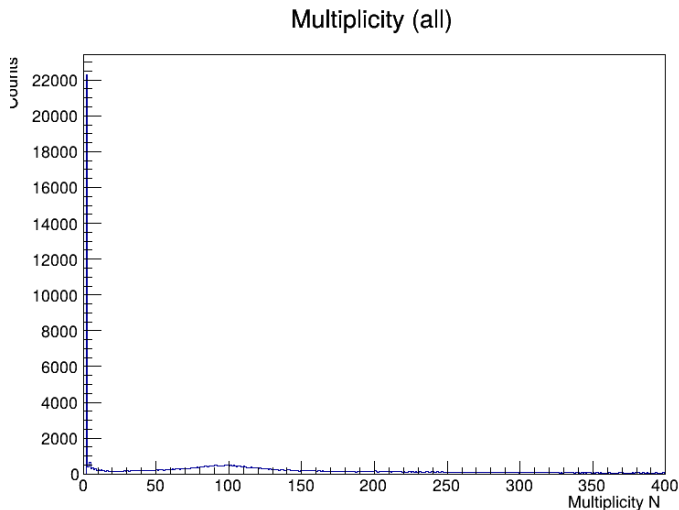
February 9, 2026

Project overview

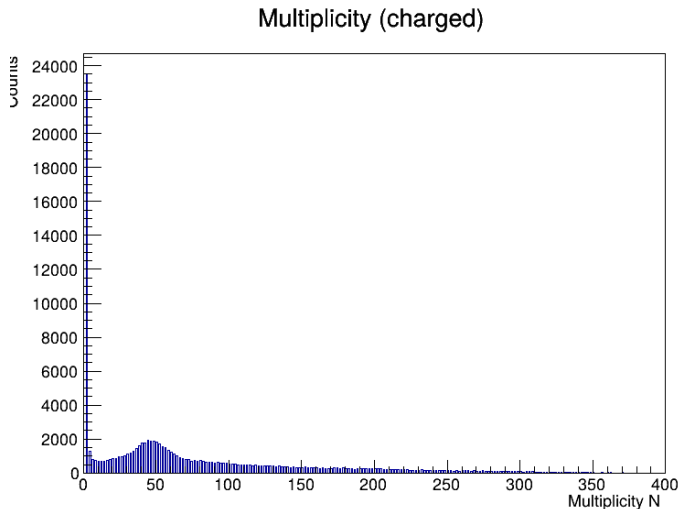
- Simulation of proton–proton collisions at $\sqrt{s} = 13.6$ TeV
- Event generation using **Pythia 8**
- Analysis of final-state particles with **ROOT**
- Distributions of N , p_T , η , and ϕ

Event-by-event particle multiplicity

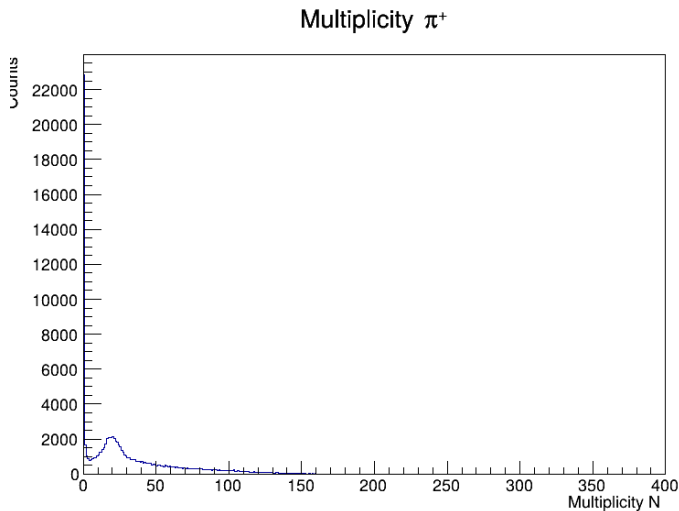
Multiplicity distribution (all particles)



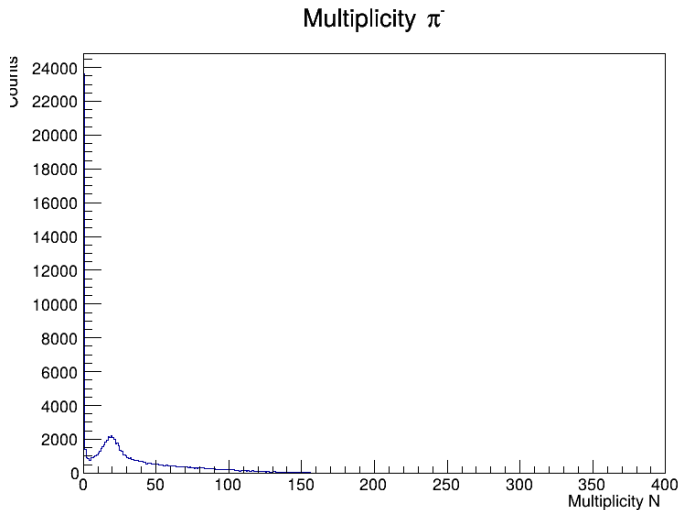
Multiplicity distribution (charged particles)



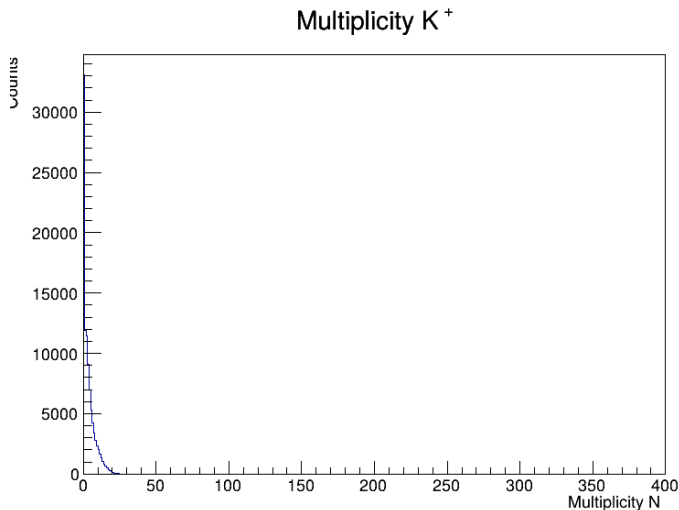
Multiplicity distribution for π^+



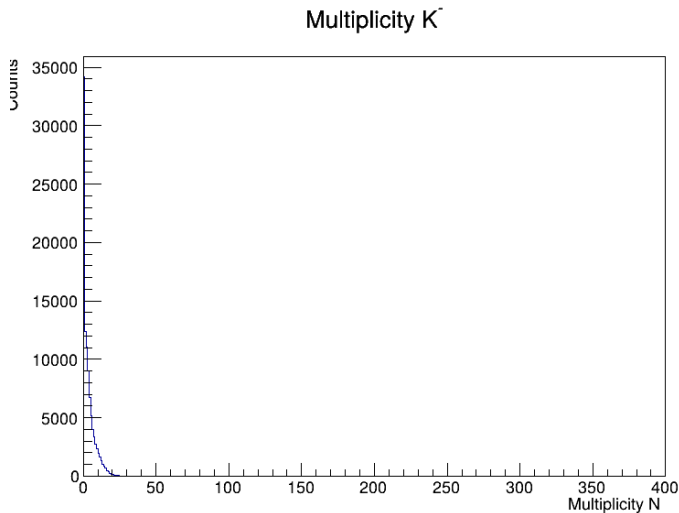
Multiplicity distribution for π^-



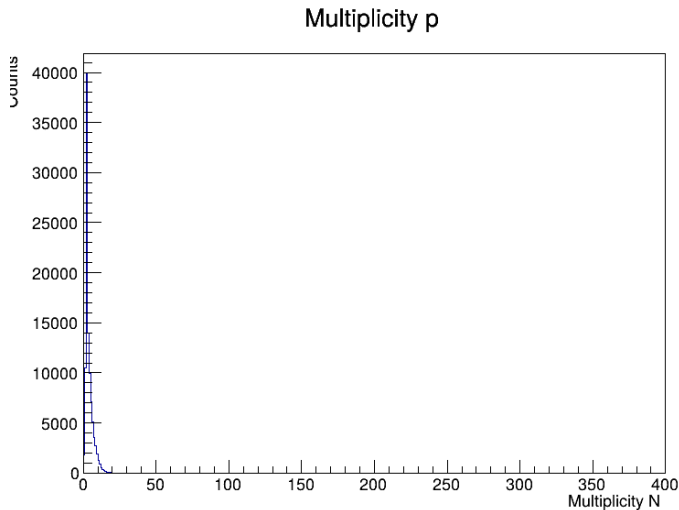
Multiplicity distribution for K^+



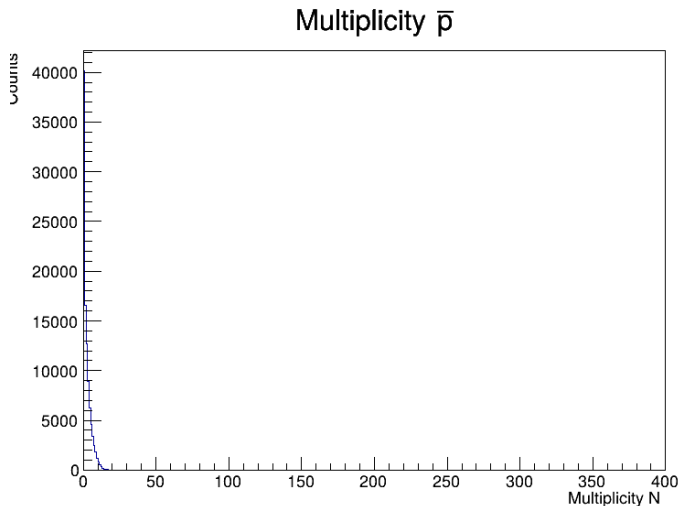
Multiplicity distribution for K^-



Multiplicity distribution for p

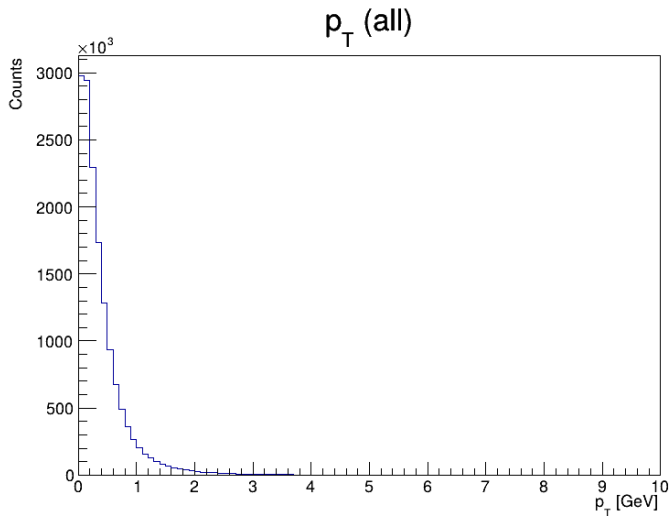


Multiplicity distribution for $\bar{p}$

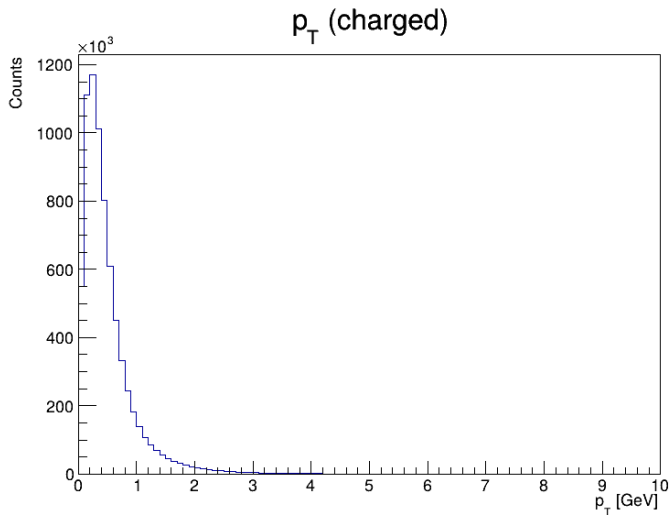


Transverse momentum distributions

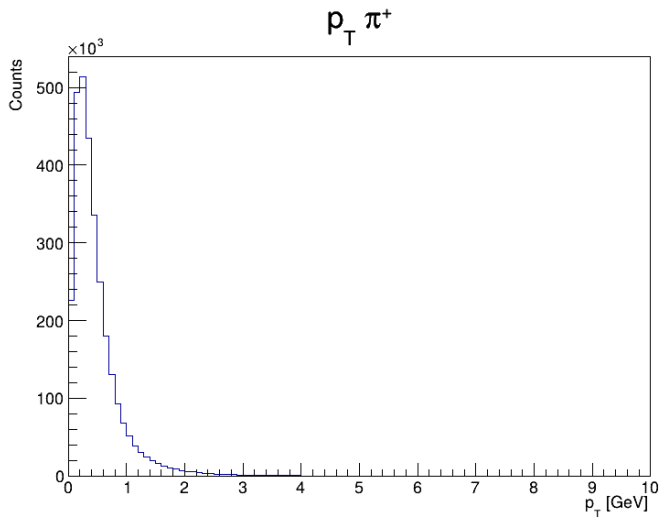
p_T distribution (all particles)



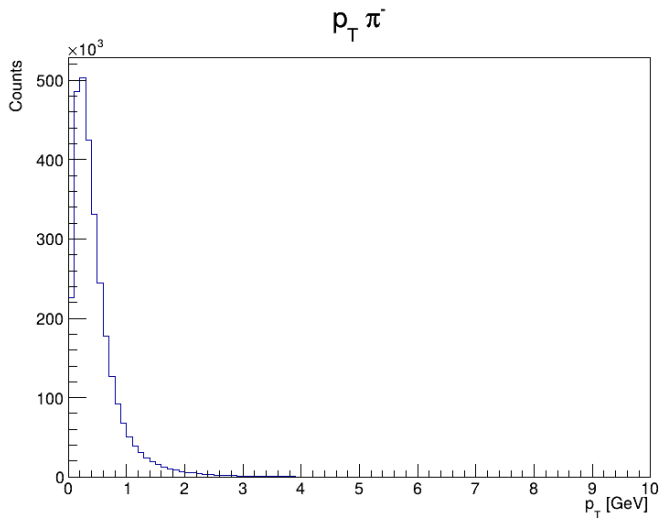
p_T distribution (charged particles)



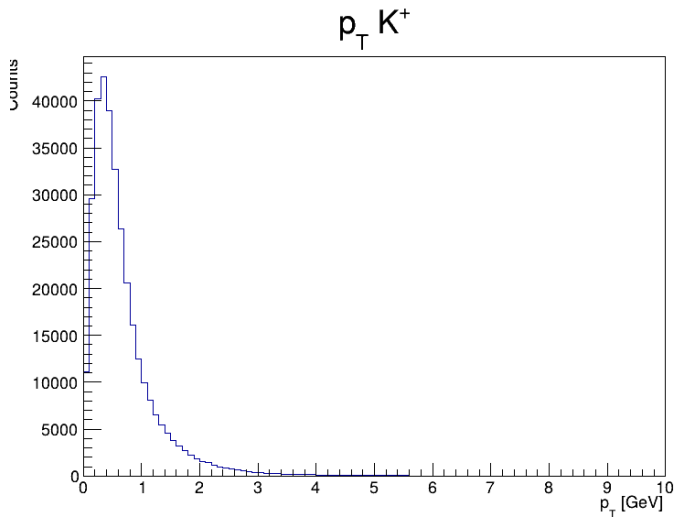
p_T distribution for π^+



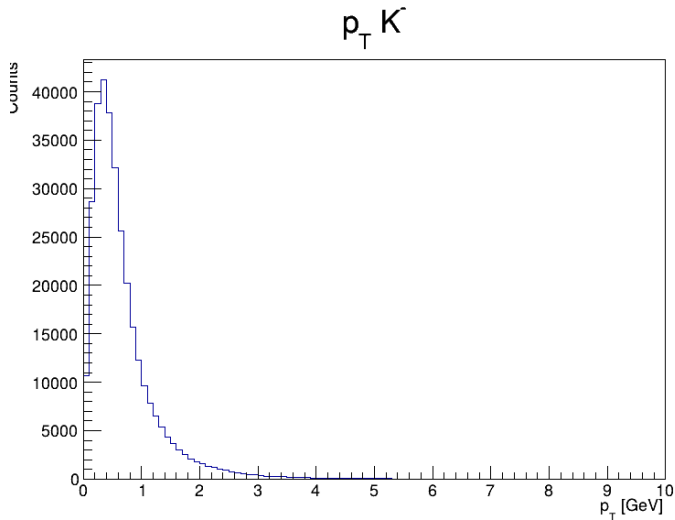
p_T distribution for π^-



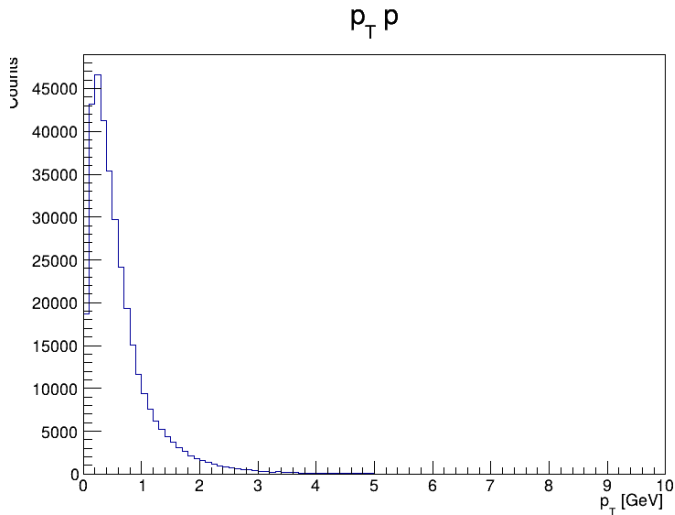
p_T distribution for K^+



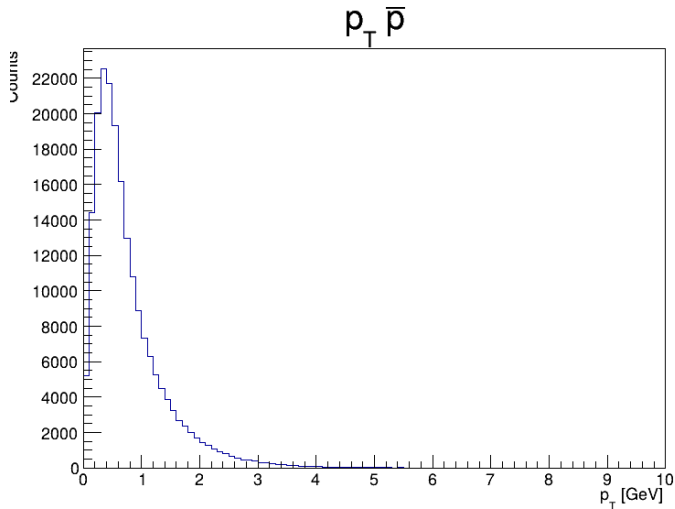
p_T distribution for K^-



p_T distribution for p

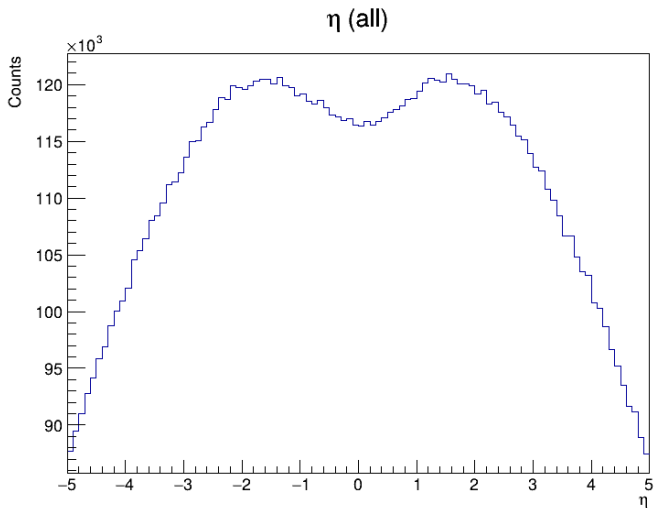


p_T distribution for $\bar{p}$

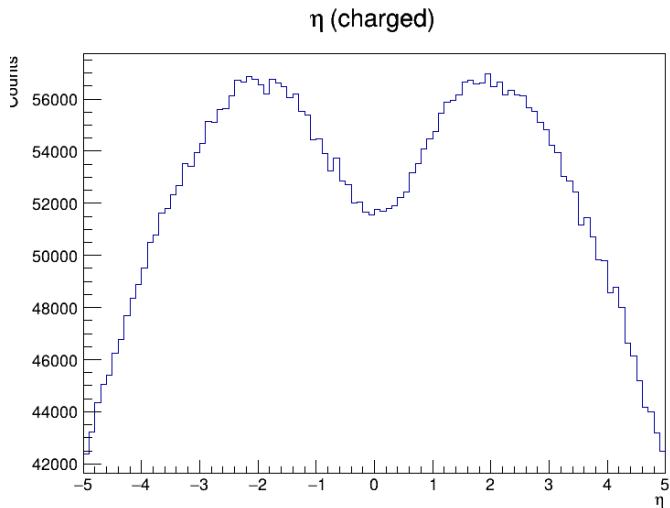


Pseudorapidity distributions

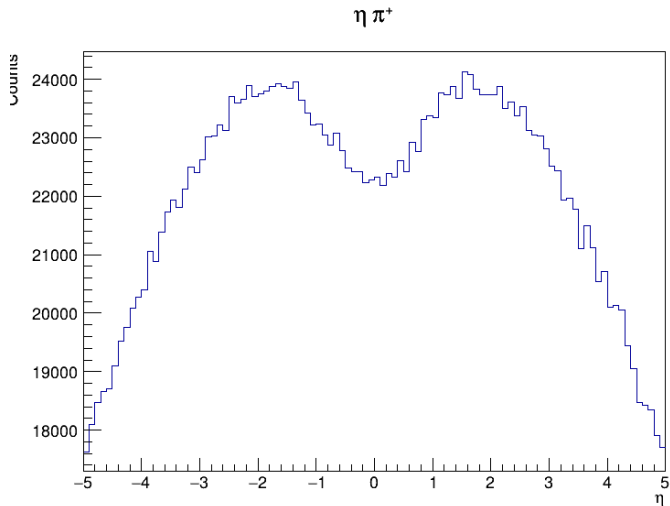
η distribution (all particles)



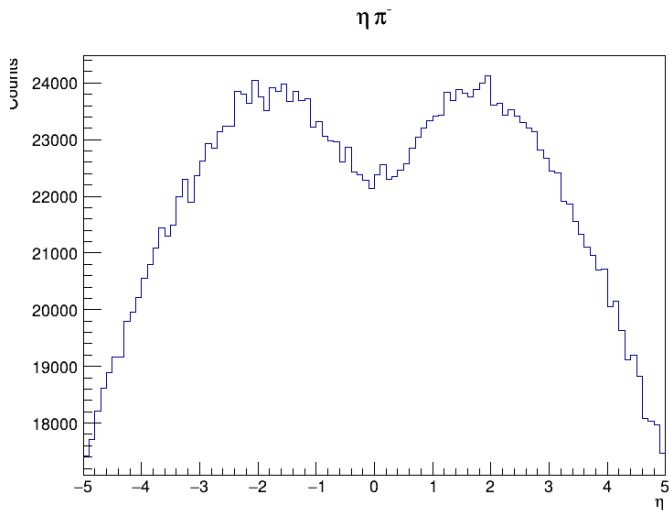
η distribution (charged particles)



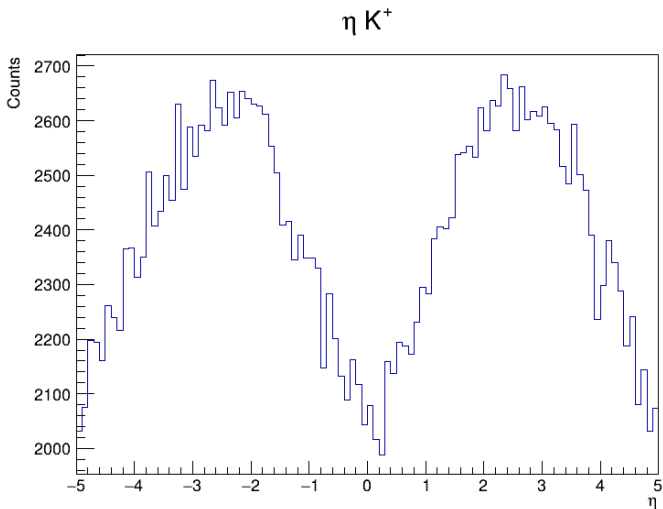
η distribution for π^+



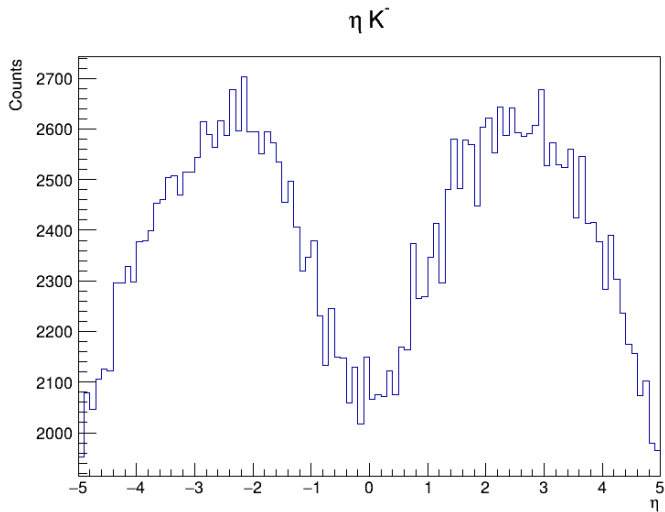
η distribution for π^-



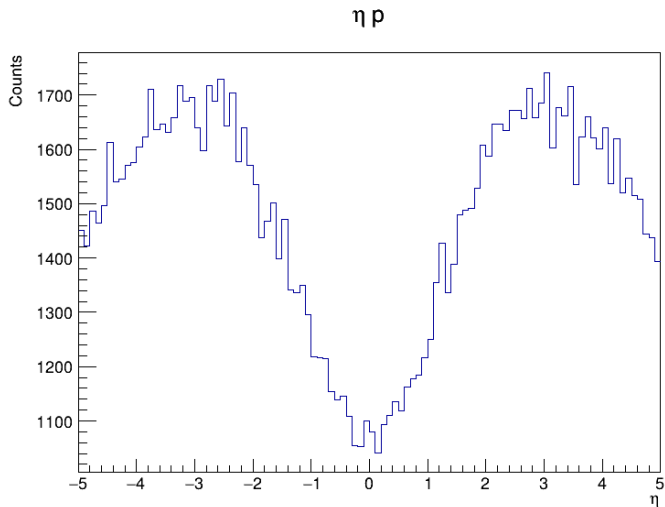
η distribution for K^+



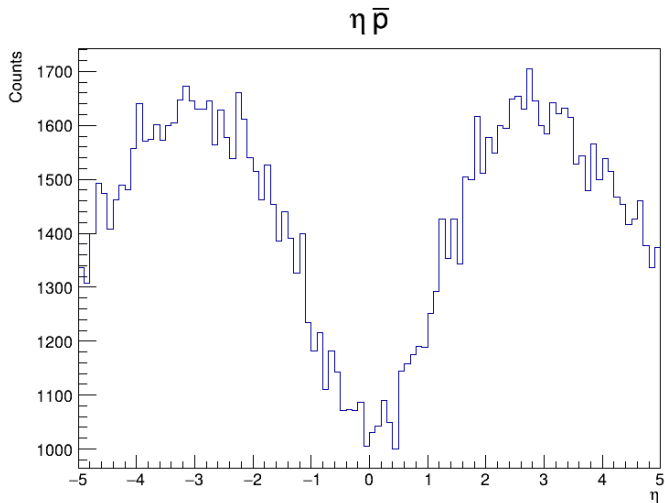
η distribution for K^-



η distribution for p

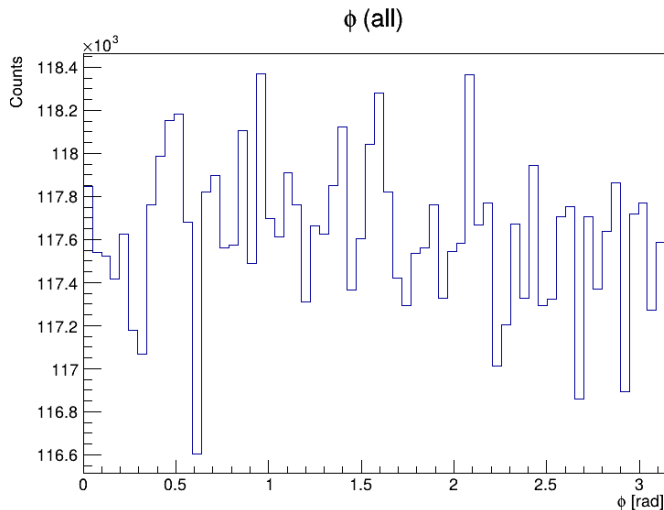


η distribution for $\bar{p}$

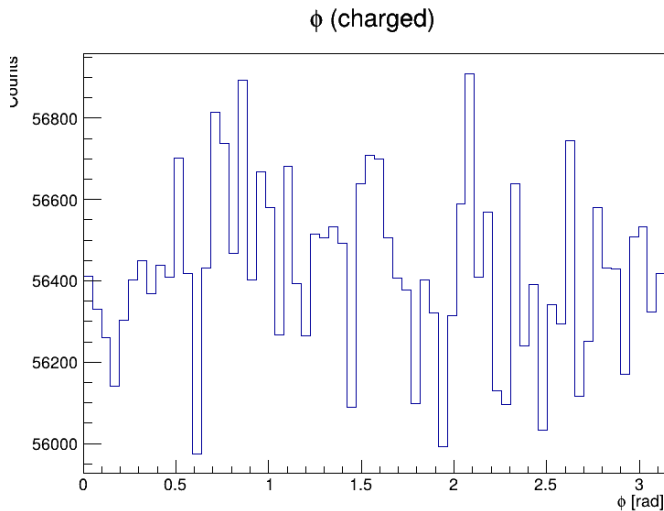


Azimuthal angle distributions

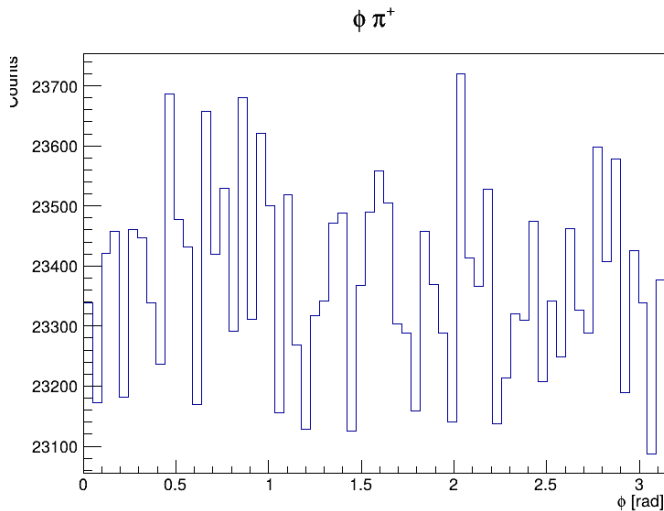
ϕ distribution (all particles)



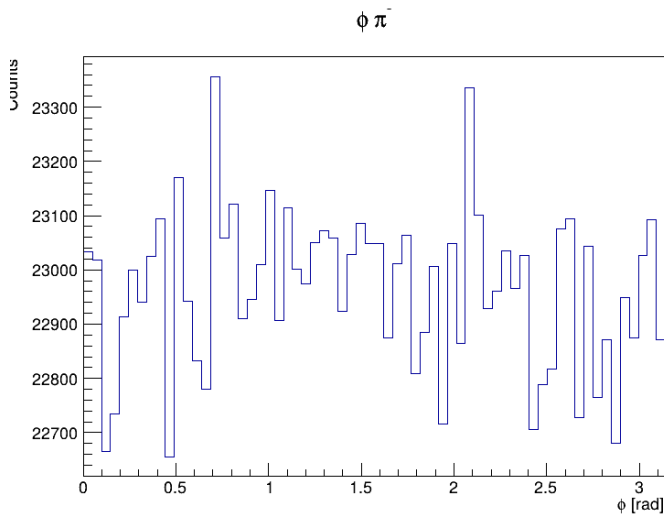
ϕ distribution (charged particles)



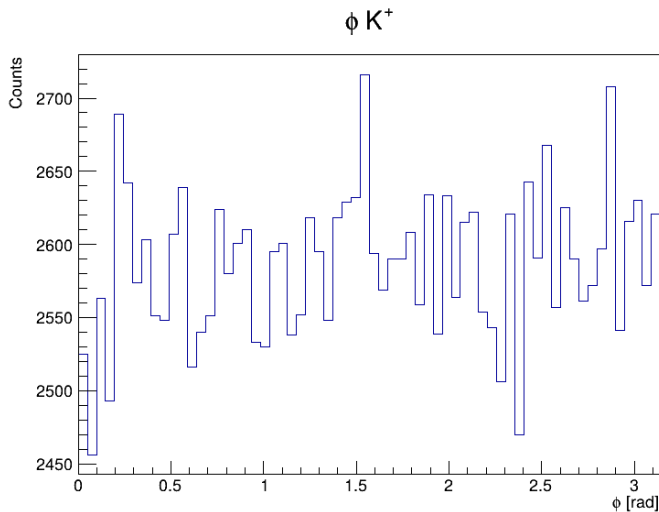
ϕ distribution for π^+



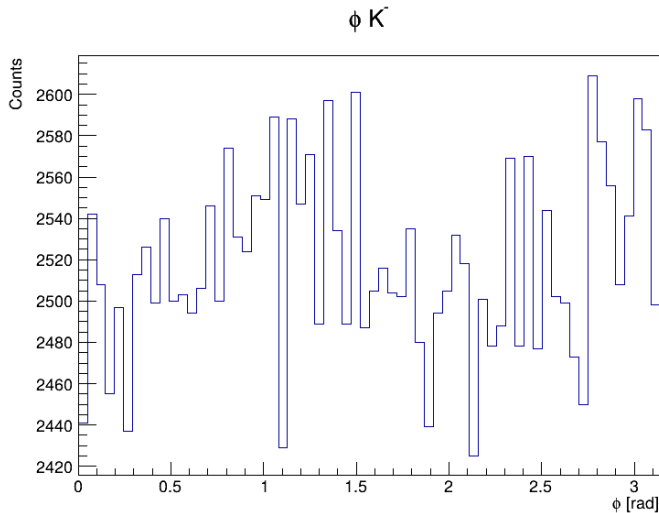
ϕ distribution for π^-



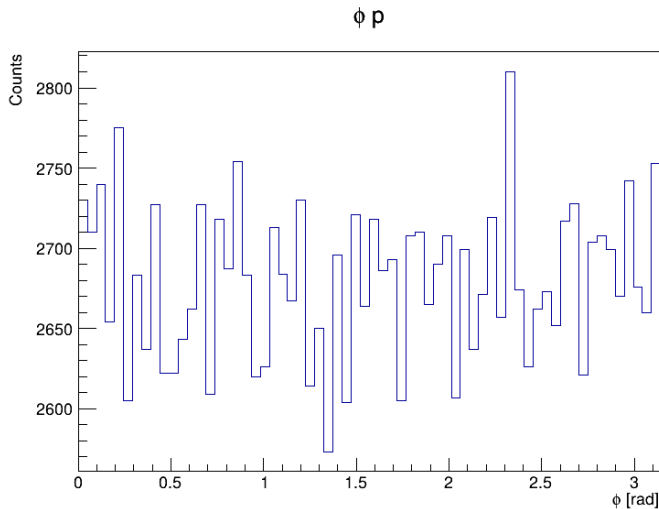
ϕ distribution for K^+



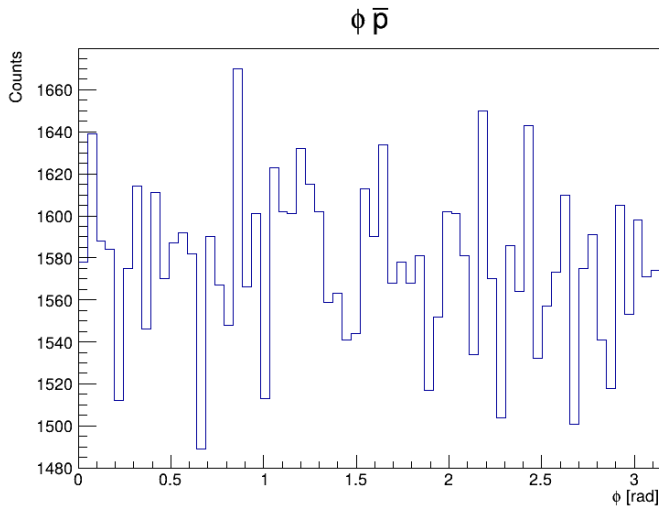
ϕ distribution for K^-



ϕ distribution for p



ϕ distribution for $\bar{p}$



Summary

- Complete analysis pipeline implemented
- Results stored in ROOT and exported to PDF/PNG
- Framework ready for further physics studies